Swiss Institute of Bioinformatics

PR STATS

Metagenomic Data Analysis of Microbial Communities (MGMC01)

Day 2 - Session 1

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Swiss Institute of Bioinformatics

PR STATS

Acknowledgment

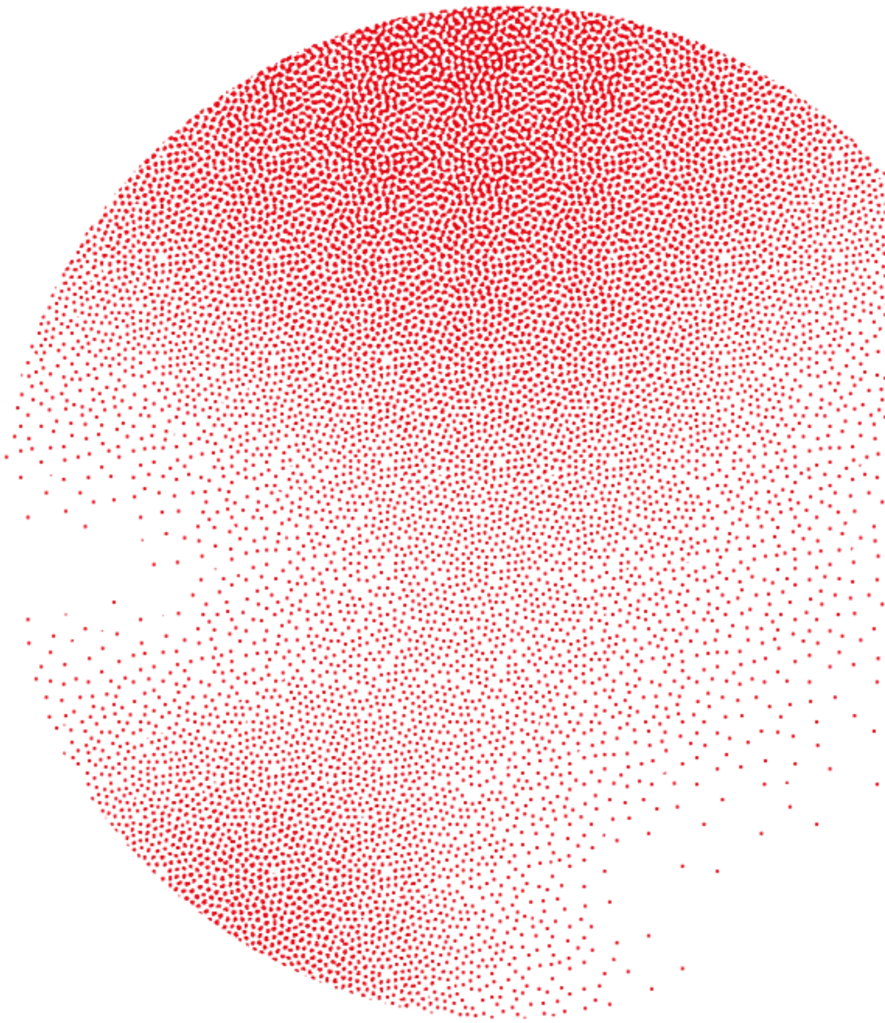


Dr. Laurent Falquet
University of Fribourg

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This presentation

- Learning outcomes
- Why Amplicon sequecing
- OTUs vs ASVs
- Pipelines
- Taxonomic classification
- Databases
- Mock communities
- Summary



Learning outcomes

- Recognize why and when amplicon sequecing is good methodological option
- Explain the principles of amplicon-based (targeted) metagenomics and the rationale behind marker gene selection (16S rRNA, ITS).
- Contrast OTU- and ASV-based workflows and justify the choice between them.
- Understand how pipelines, like DADA2, process amplicon reads to generate ASVs.

The choice of the method depends on 3 main questions

Who is there?

catalogue of species, diversity,genus, etc.

distribution (how many of each)

What are they doing?

genes/proteins, GO terms, metabolic pathways

functional annotation, bioprospection

How do they compare?

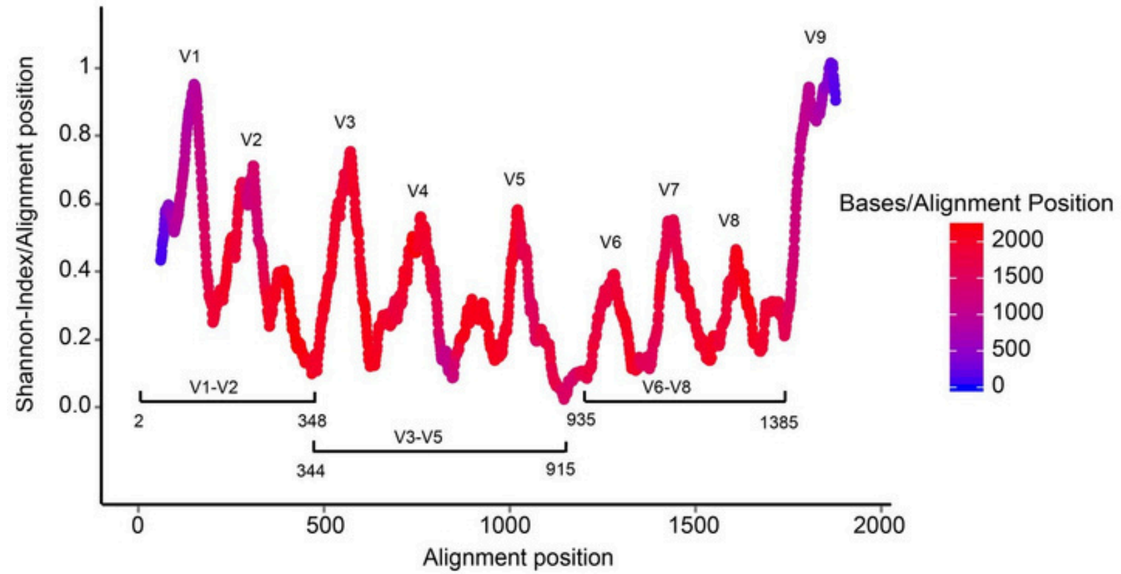
pairwise or multiple comparisons

correlation with environmental factors



Reminder about 16S rRNA variable regions

Usually this is a gene with some variable sequences in all species surrounded by conserved sequences:



Why Amplicon sequencing?

Targeted sequencing of marker genes (16S, ITS)

01 ·

Cost-effective

Many samples at low cost per sample

02 ·

High sensitivity

Works with low-biomass and low-input samples

03 ·

Community profiling

Who is there: composition and diversity

04 ·

Little host DNA

Targeted PCR avoids host and off-target DNA

05 ·

Established workflows

Mature pipelines and curated reference databases

06 ·

Light computing

Small datasets, analysis on a standard laptop

Targeted or Amplicon sequencing

Library preparation requires PCR amplification

QC (Clean reads)

optional: combine overlapping PE

Align reads to reference database

Cluster reads to OTUs/ASVs

Assign OTUs/ASVs to taxonomy

Tools (examples):

Mothur

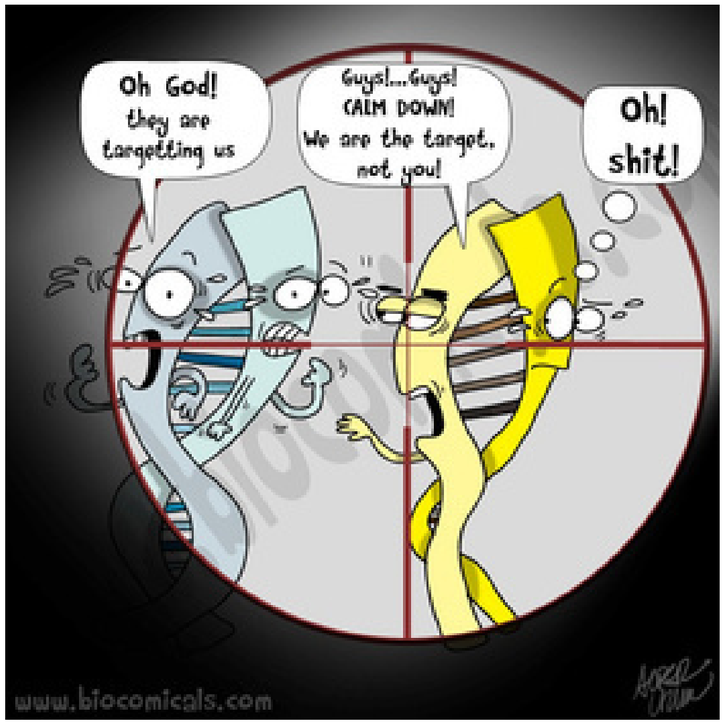
Qiime2

DADA2

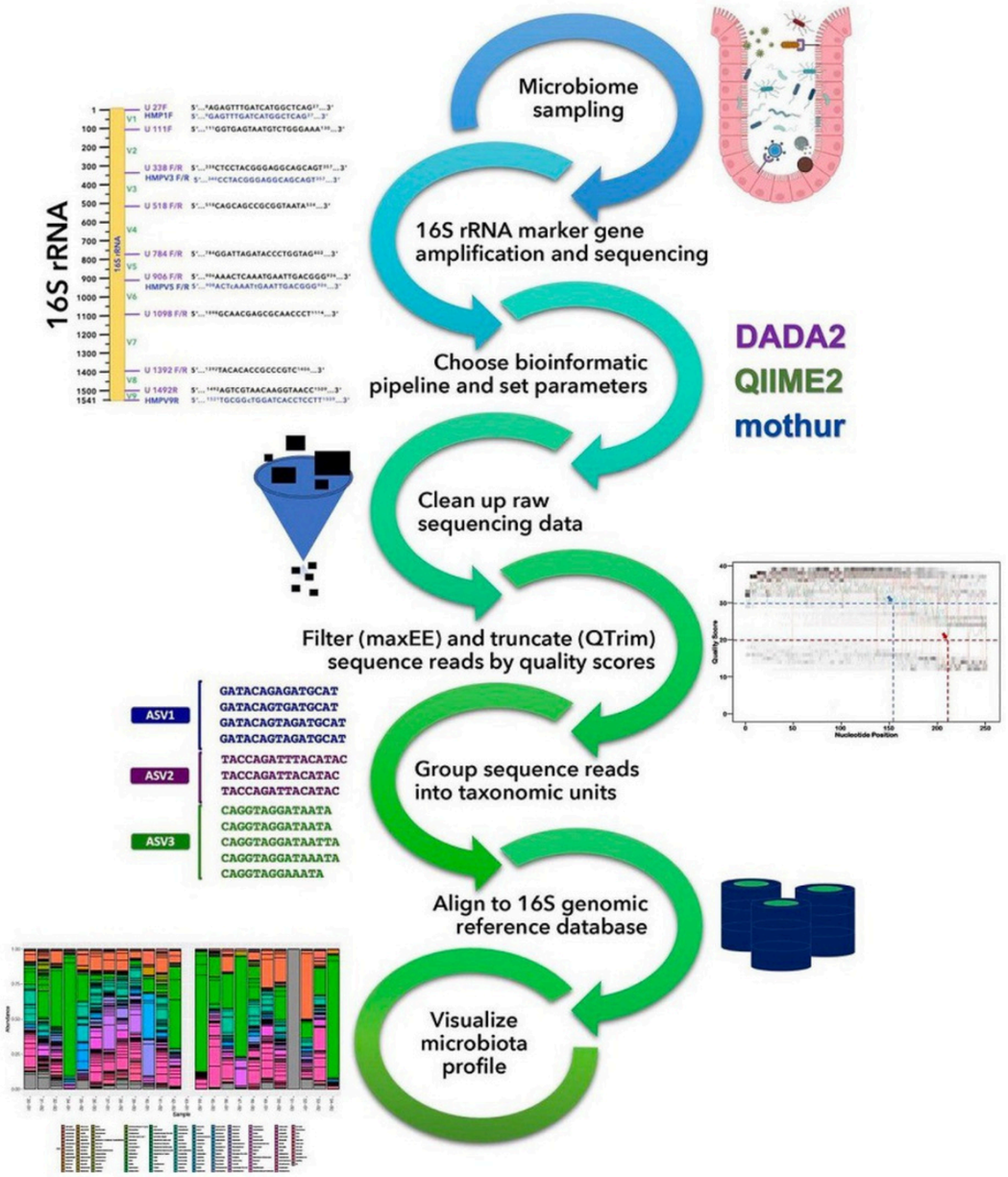
MEGAN

MetaPhlAn4

PIPITS (for fungi)



The typical pipeline



OTUs vs ASVs

OTU = Operational Taxonomic Unit
ASV = Amplicon Sequence Variant

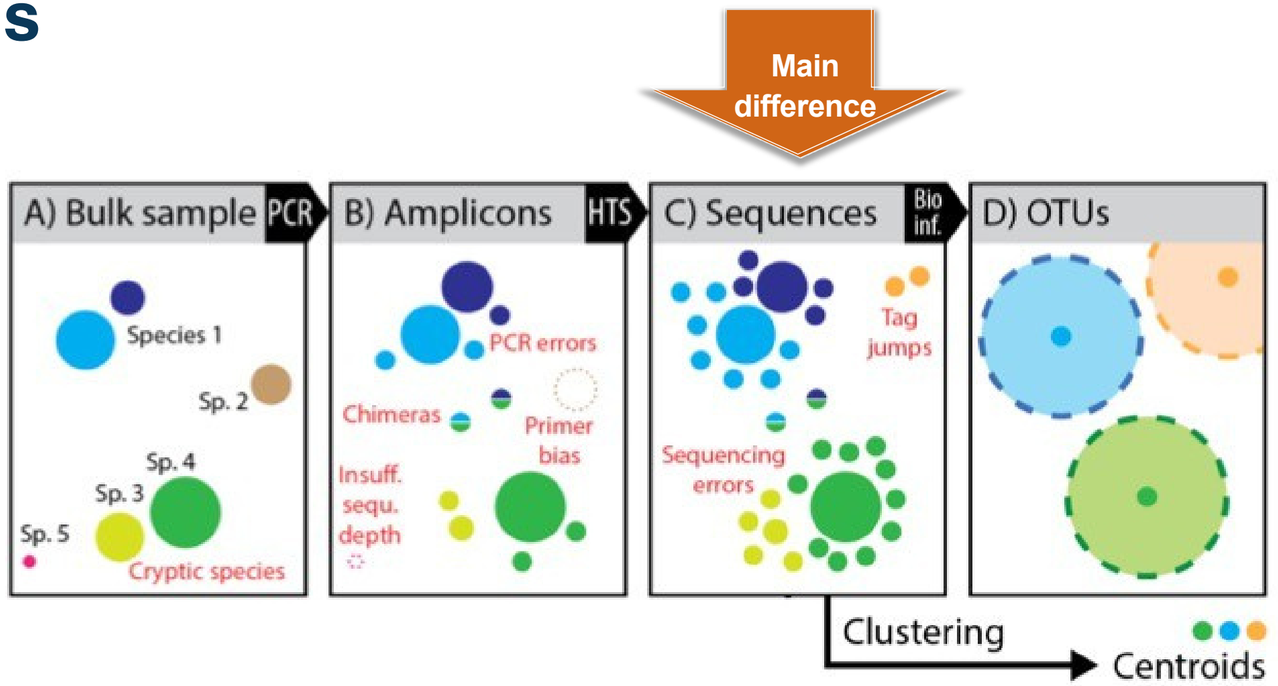
Both are *in silico* classifications of the PCR amplicons with two opposite philosophies:

OTUs are clustered sequences to centroids
ASVs are denoised sequences to haplotypes

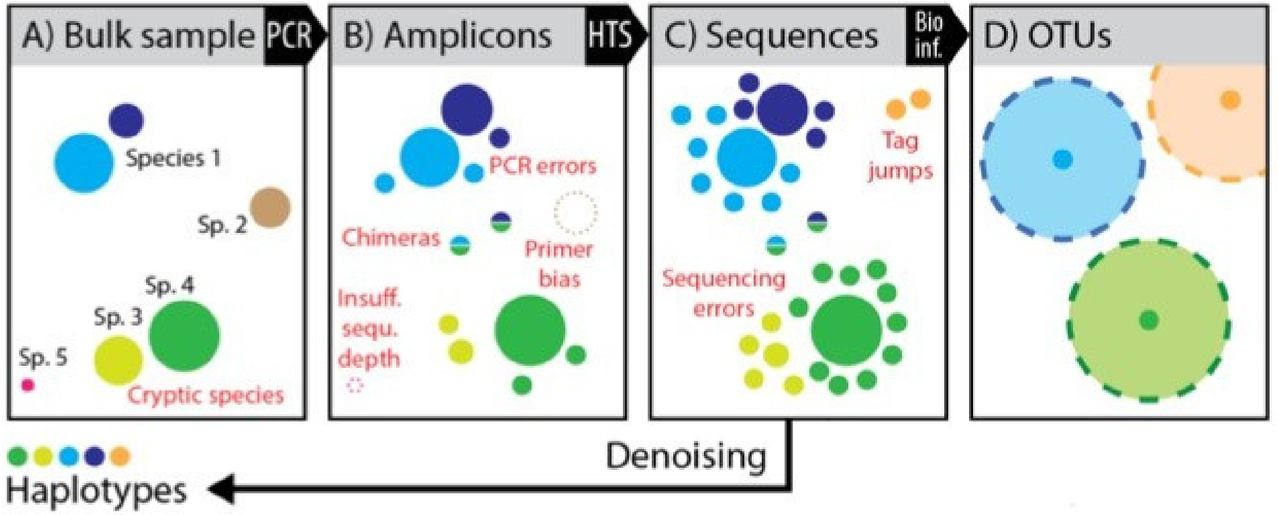
The final goal is to obtain feature tables and representative sequences

OTUs vs ASVs

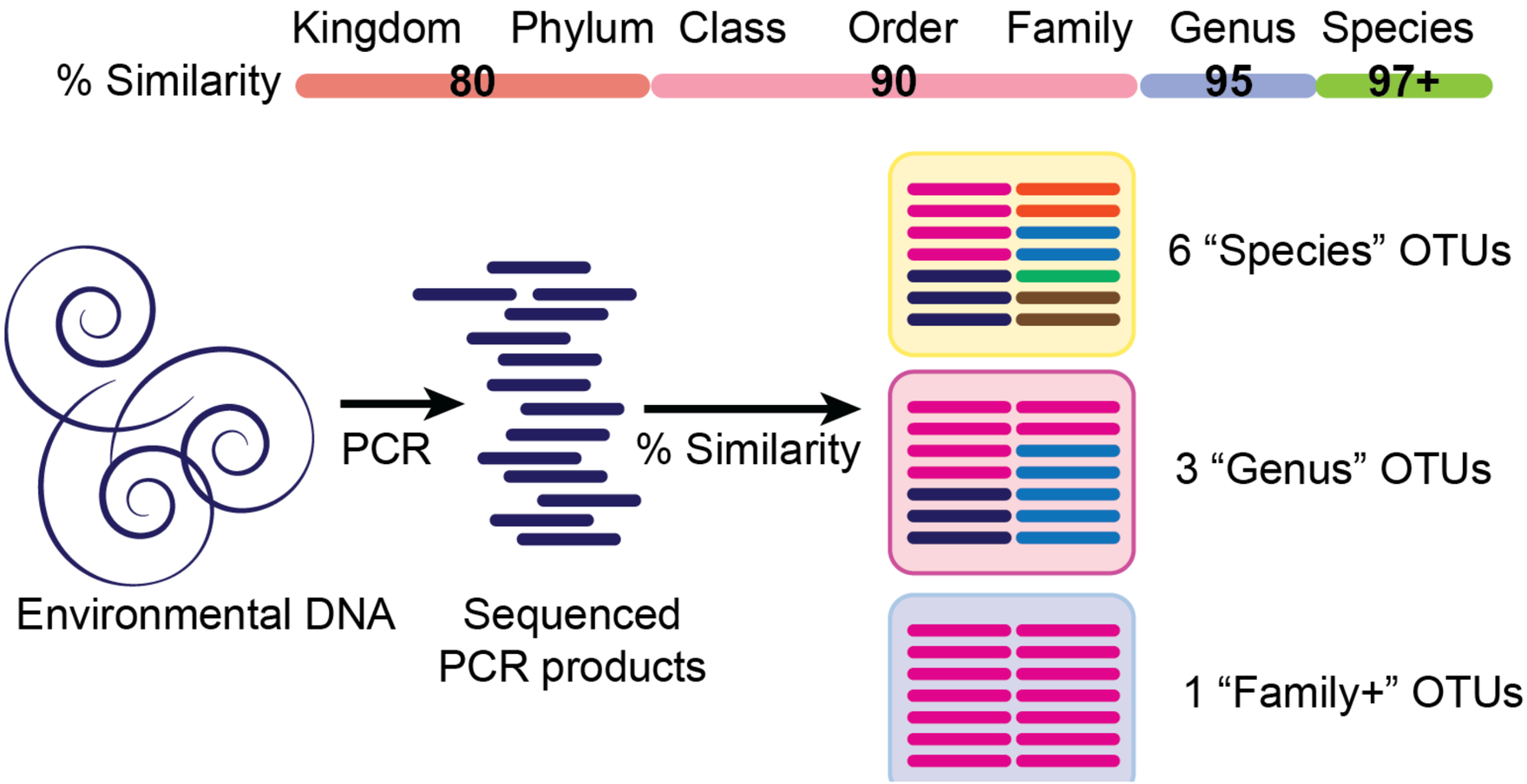
OTUs



ASVs

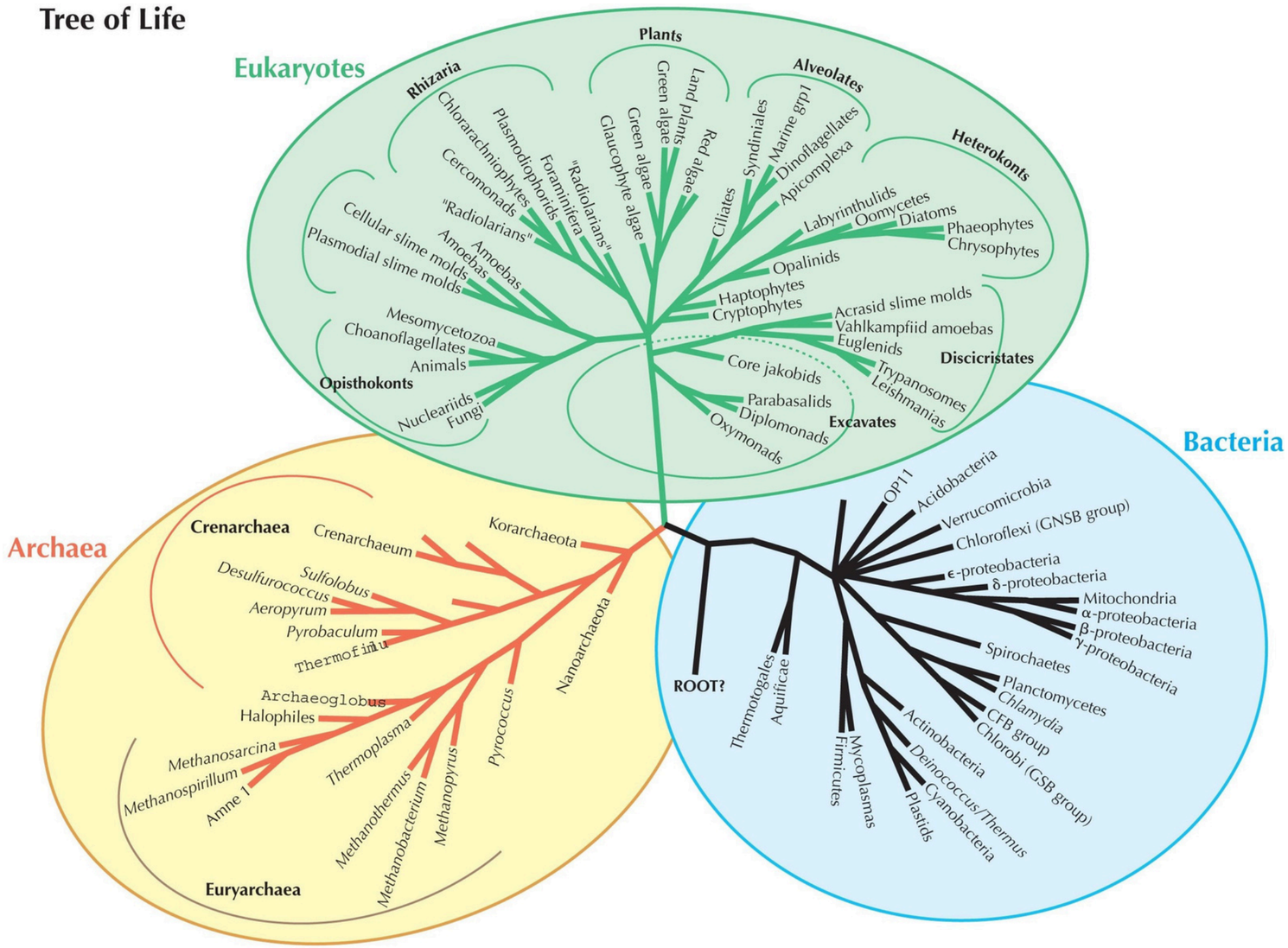


Operational Taxonomic Units

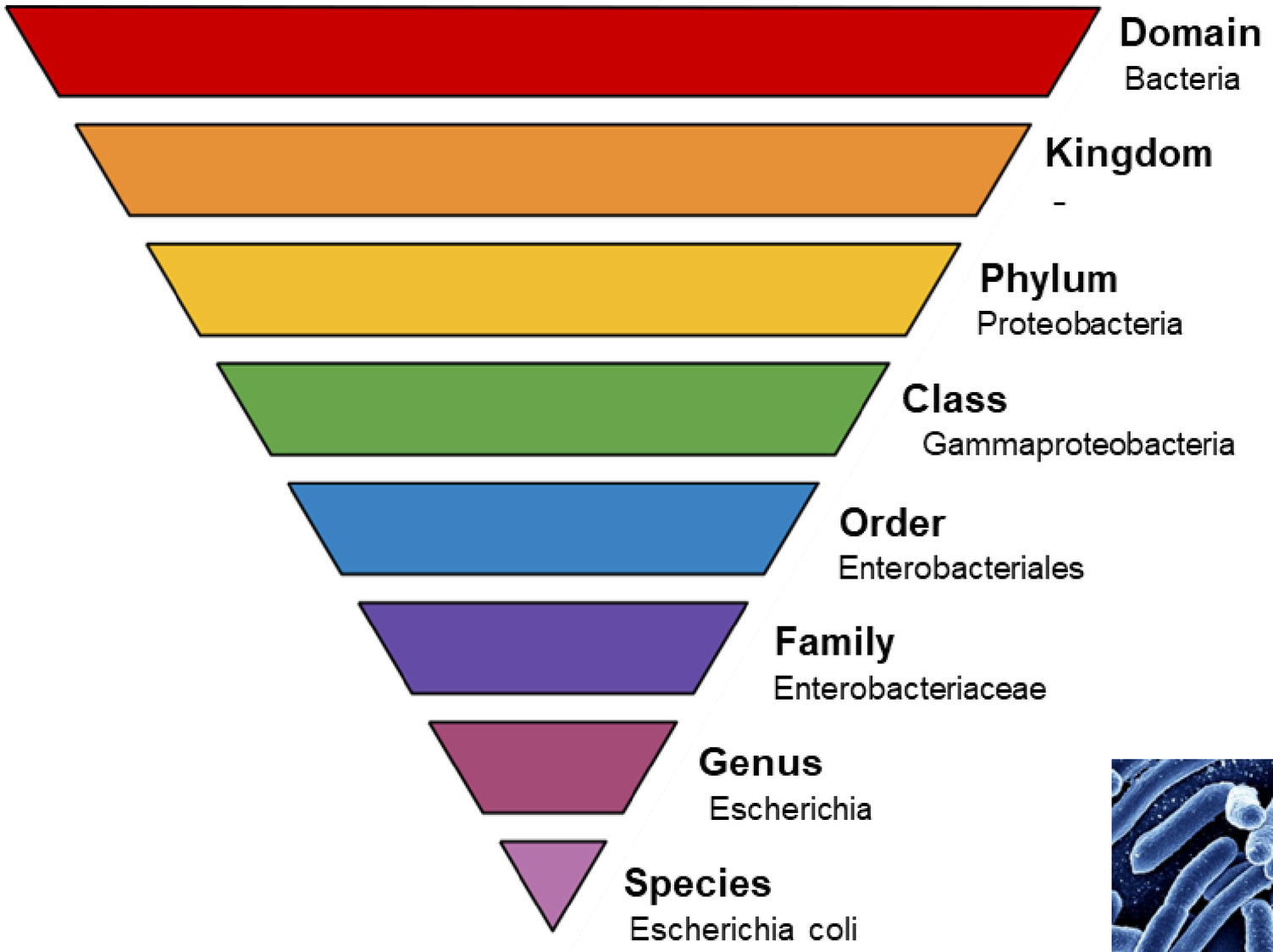


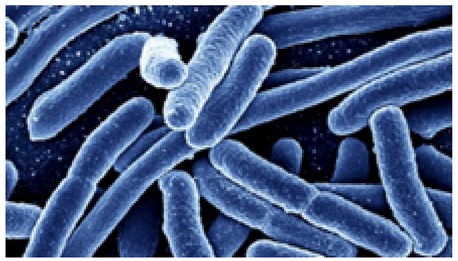
For 16S rRNA OTUs (~species) are delineated with a 3% sequence dissimilarity and higher taxa with increasingly larger dissimilarity

The tree of life (genealogy of the living organisms)

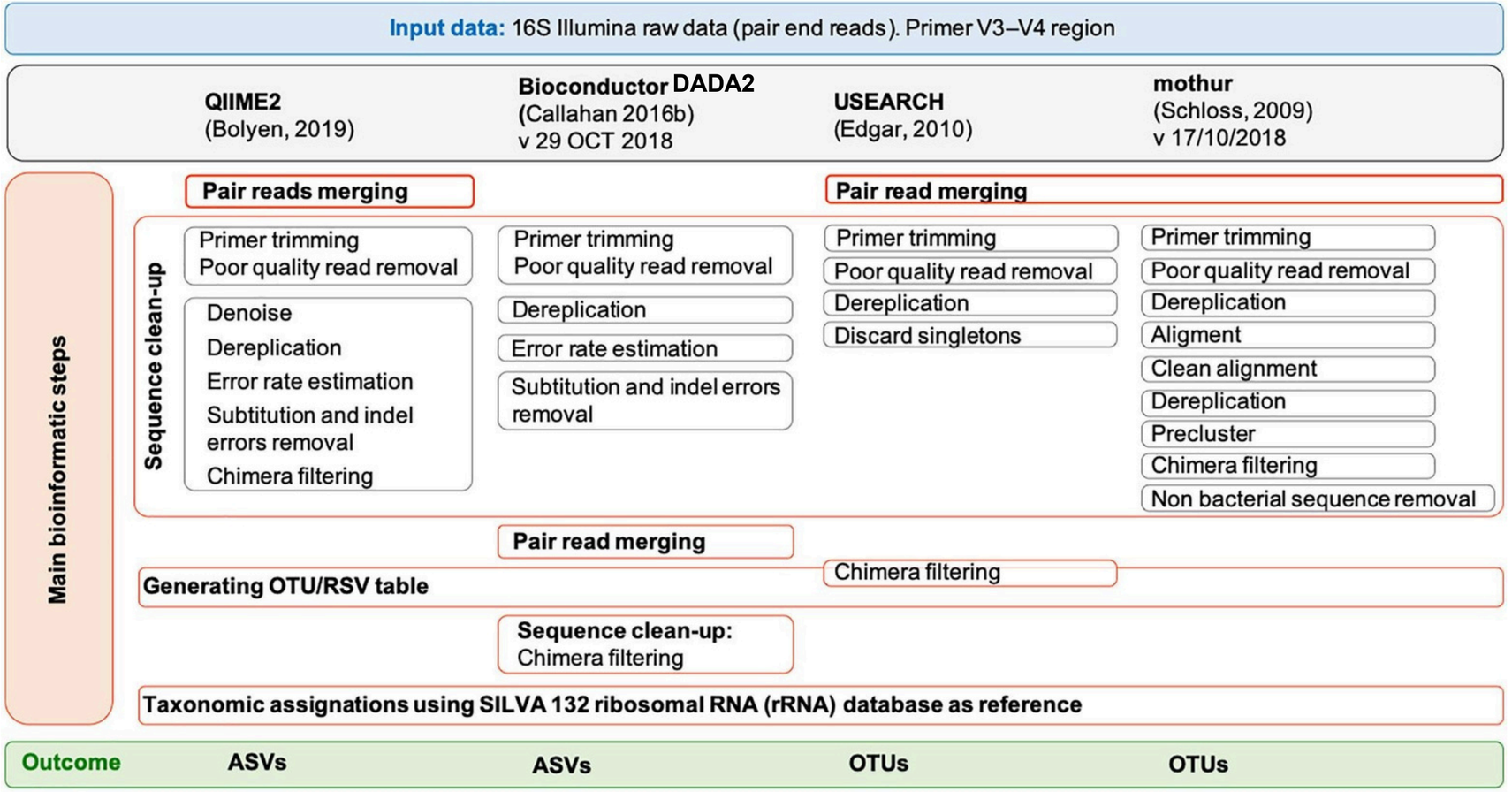


Taxonomy of bacteria?





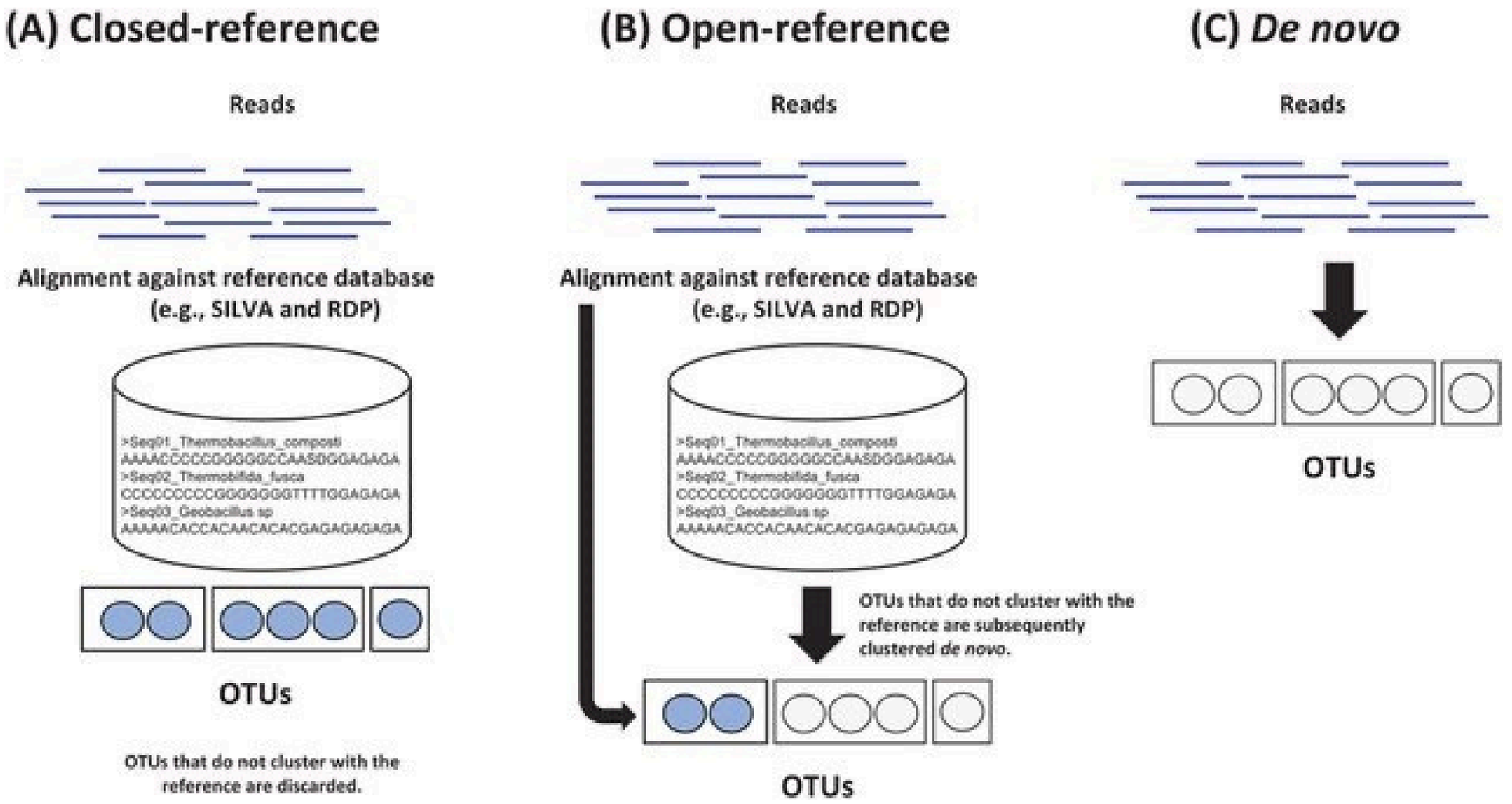
Comparison of pipelines



	QIIME2	Bioconductor DADA2	UPARSE	Mothur
Open source	√ Yes	√ Yes	Ø **32-bit only**	√ Yes
Installation	√ Clear online instruction Ø Specific version dependency[a]	√Ø Easy for R user	√ Installation not required	√ Installation not required
User support	√ Tutorials, forum, SOPs	√ Tutorials, forum, SOPs	√ Video tutorials, forum, SOPs	√ Tutorials, forum, SOPs
Specific computer skills	√ Not required	Ø Required	√ Not required	Ø Required

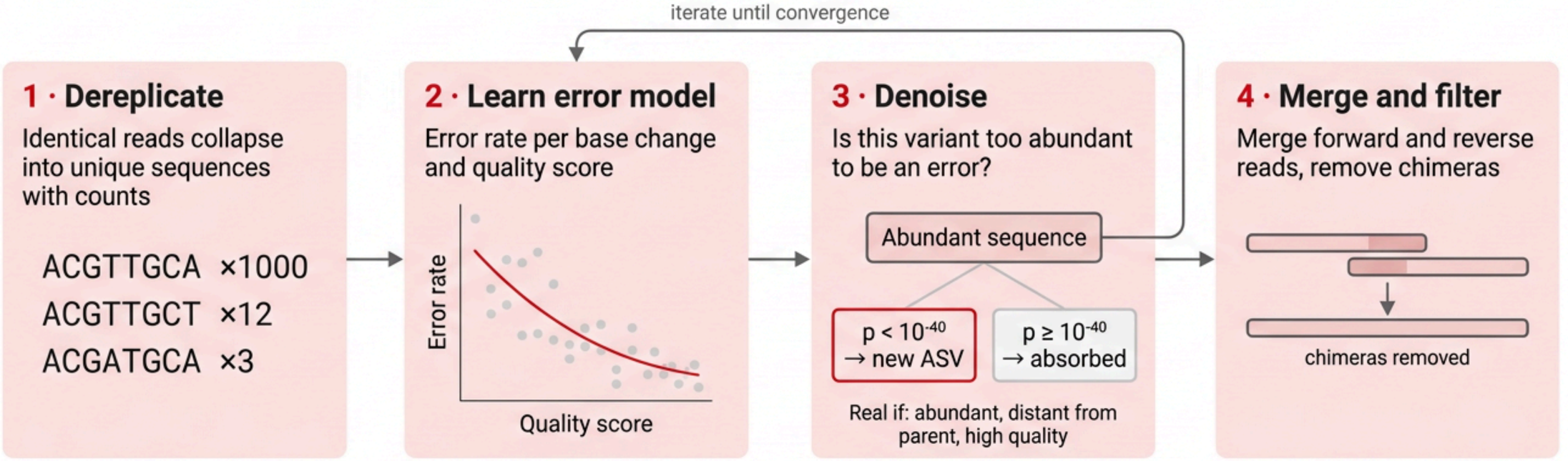
[a]*Marks indicate pros and cons, respectively. "Ø" and "√" Miniconda 2 or 3 depending on python 2.7 or 3.7, respectively.*

De novo vs Closed vs Open OTU taxonomy assignment



How DADA2 infers ASVs

Denoising with an error model learned from your own data



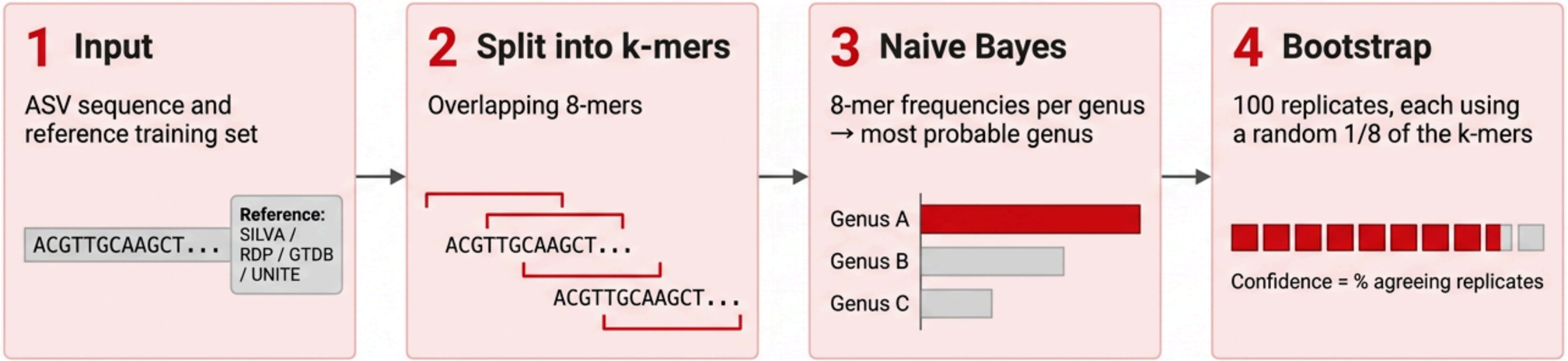
Output: ASV table

ASV	S1	S2	S3
ASV1	120	0	45
ASV2	8	63	0
ASV3	0	17	210

Taxonomy assignment in DADA2

How assignTaxonomy() works

RDP naive Bayesian classifier · k-mer size 8 · 100 bootstraps



Output (minBoot = 50)

Rank	Assignment	Bootstrap
Phylum	Firmicutes	100
Family	Lactobacillaceae	97
Genus	Lactobacillus	88
Species	NA	below 50

Feature tables and representative sequences

The list of OTUs or ASVs and read counts associated to a taxon

OTU	Size	Taxonomy
Otu00001	530053	Bacteria(100);Proteobacteria(100);Gammaproteobacteria(100);Pseudomonadales(100);Pseudomonadaceae(100);Pseudomonas(100);
Otu00002	298855	Bacteria(100);Proteobacteria(100);Betaproteobacteria(100);Methylophilales(100);Methylophilaceae(100);Methylophilus(90);
Otu00004	269985	Bacteria(100);Proteobacteria(100);Gammaproteobacteria(100);Pseudomonadales(100);Pseudomonadaceae(100);Pseudomonas(99);
Otu00006	100666	Bacteria(100);Proteobacteria(100);Betaproteobacteria(100);Burkholderiales(100);Comamonadaceae(100);Variovorax(96);
...		

The list of OTUs or ASVs and read counts per sample

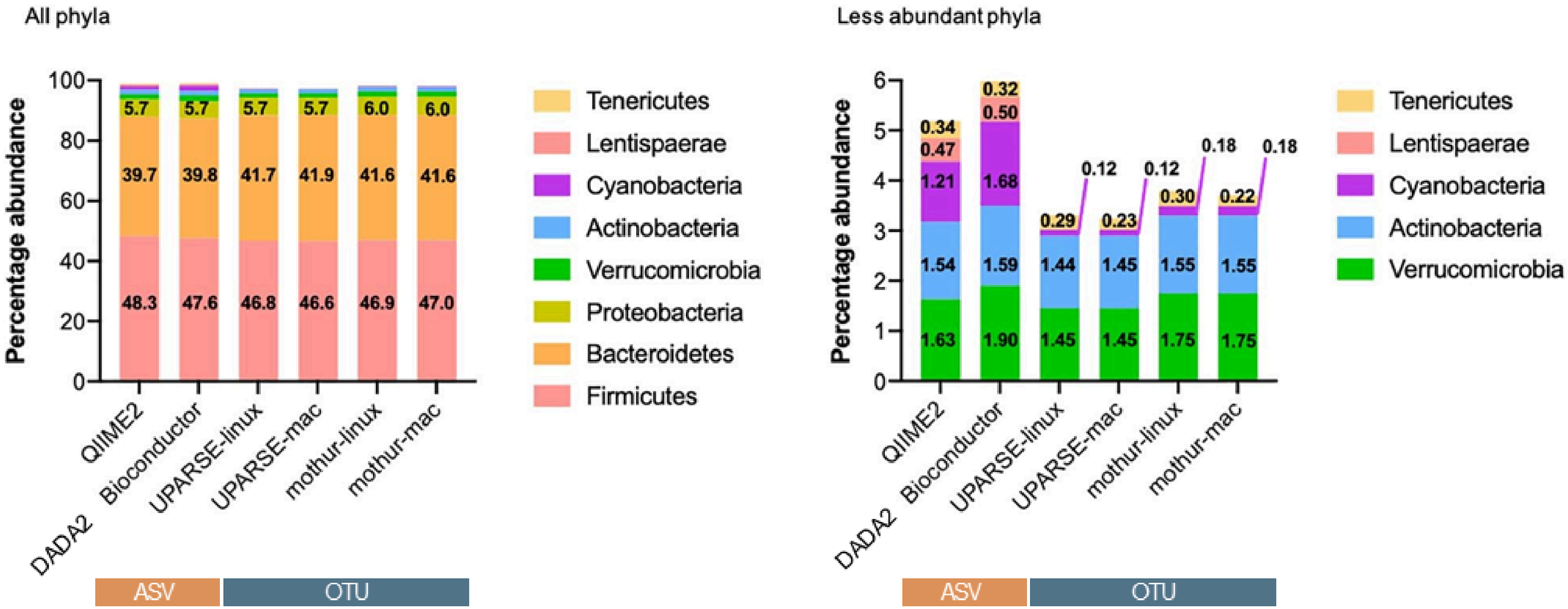
label	Group	numotus	otu00001	otu00002	otu00003	otu00004	otu00005	otu00006	otu00007	otu00008 ...
0.03	bdg-ws-1a	11867	17387	27426	11854	207	53	5808	5964	2006 ...
0.03	bdg-ws-1b	2999	28979	12	98712	87	127	17738	727	771 ...
...										

Representative sequences

Each OTU **has a single associated sequence for its cluster (centroid)**

Each ASV **is a unique sequence**

Comparison of pipelines



Marizzoni et al. (2020)

Which taxonomic database to choose?

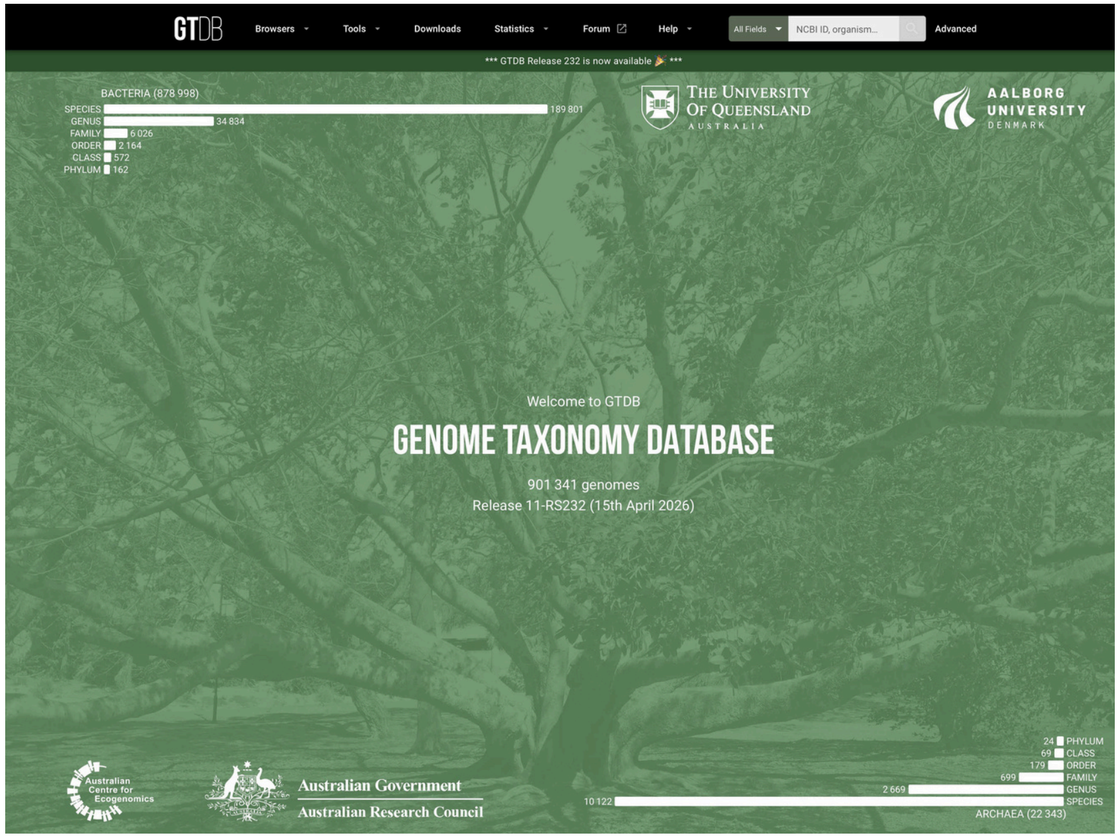
	GG2	SILVA	NCBI	RDP
URL	https://greengenes2.ucsd.edu	https://www.arb-silva.de	https://www.ncbi.nlm.nih.gov/refseq/targeted loci/	https://jfq3.gitbook.io/rdptools-docker/
Last release	2024.09	138.2 Jul. 2024	267 Oct. 2025	11.5 Sept. 2016 (outdated)
SSU sequences	23‘467‘470	2’224’690	27’732	3‘356‘809
LSU sequences	NA	227’318	11’754	125‘525

Genome Taxonomy Database

New trend GTDB (classification to species level with whole genome)

https://gtdb.ecogenomic.org/

Complete genome database, not only 16S



Advantages of ASVs vs OTUs

ASVs combine advantages of both closed-reference OTUs and *de novo* OTUs

- computational costs that scale linearly with study size
- simple comparison between independently processed data sets
- allow classification prediction
- accurate measurement of diversity
- applicability to communities not fully in reference databases

	ASVs	De novo	Closed-ref
Precise	✓	~	~
Tractable	✓	~	✓
Reproducible	✓	X	✓
Comprehensive	✓	✓	X

Inclusion of a mock community

b|e|i RESOURCES

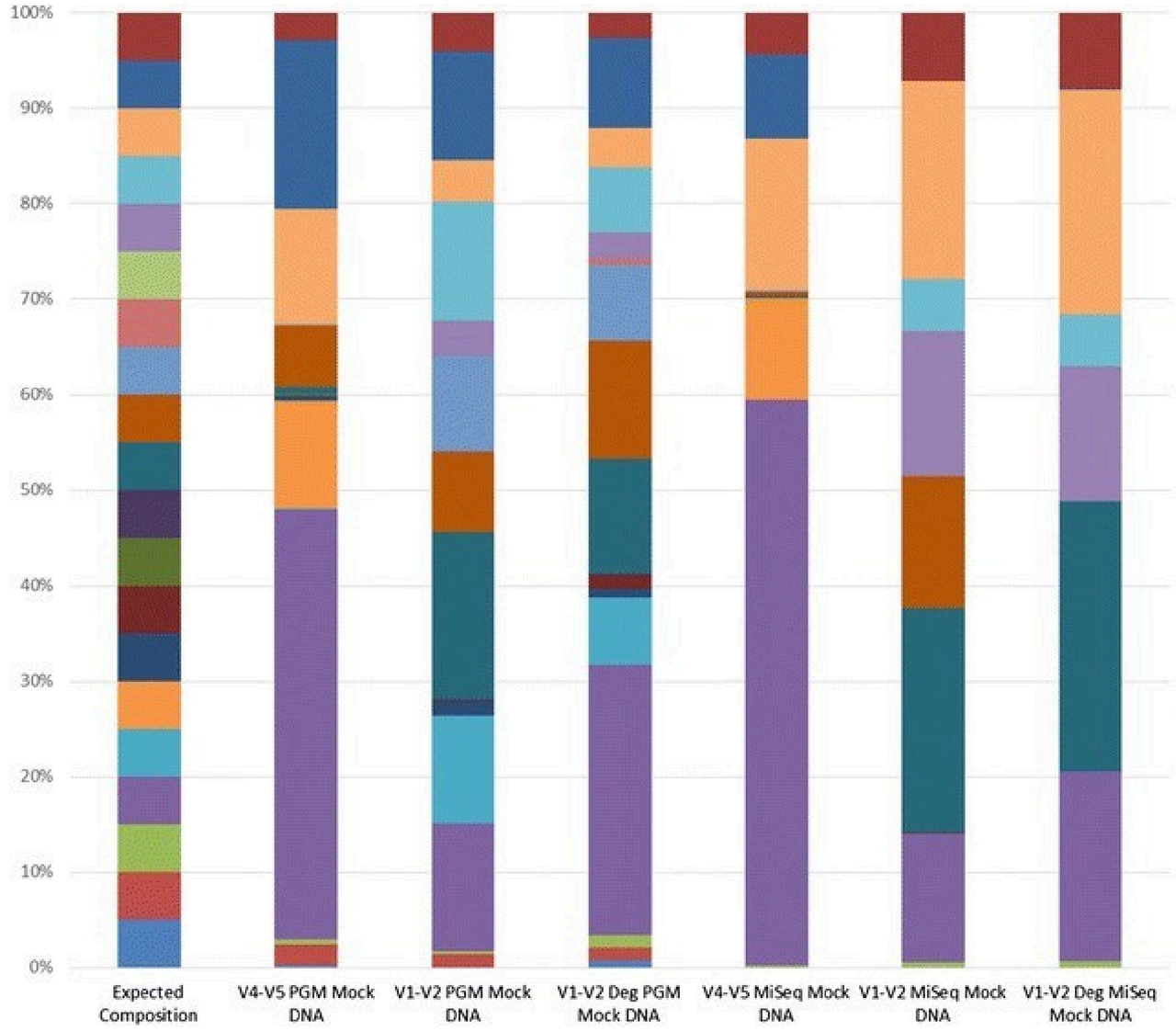
SUPPORTING INFECTIOUS DISEASE RESEARCH

Prod

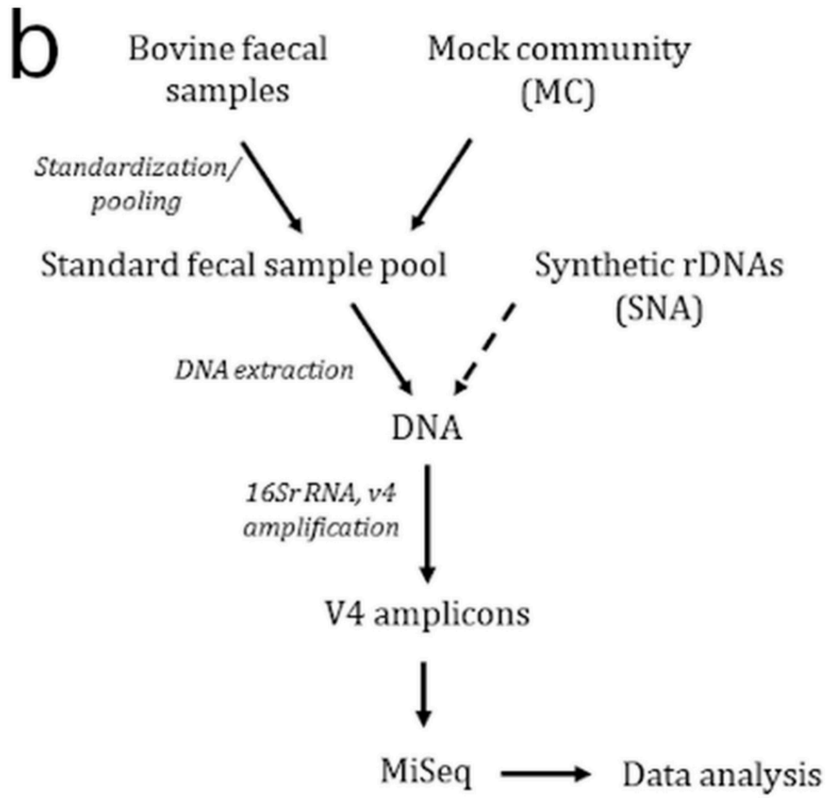
Genomic DNA from Microbial Mock Community B (Even, Low Concentration), v5.1L, for 16S rRNA Gene Sequencing

Catalog No. HM-782D



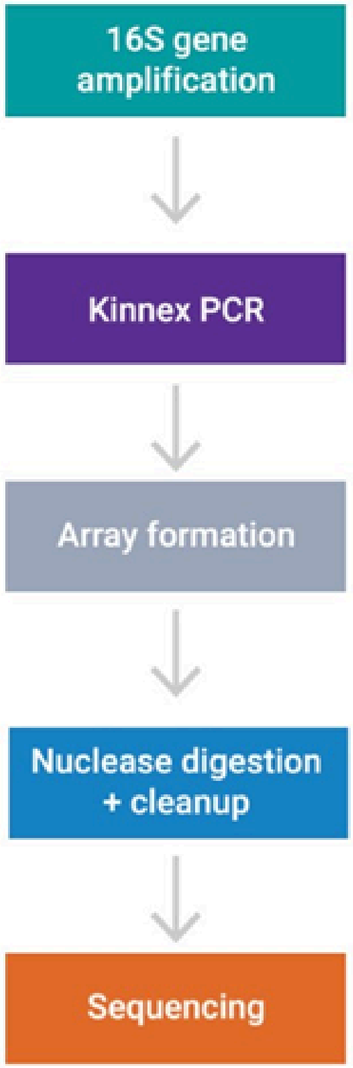


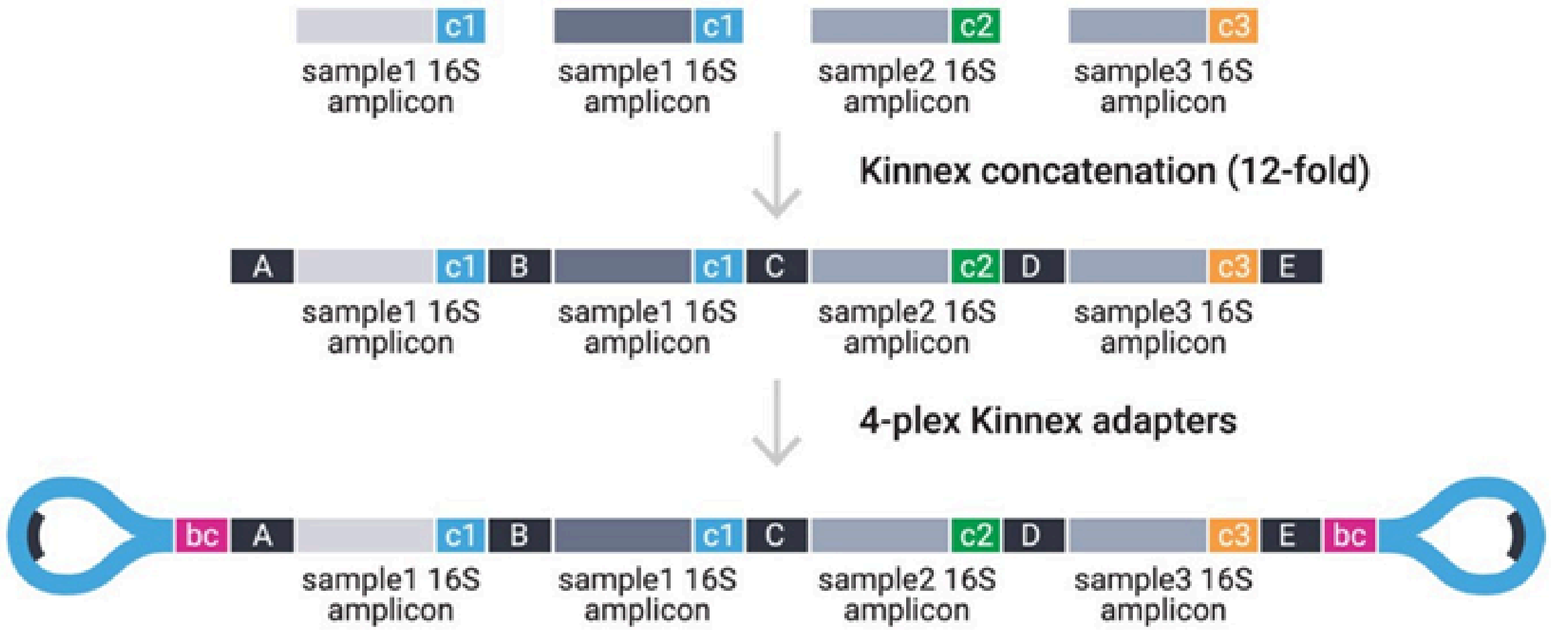
Inclusion of a mock community



Mock community as an in situ positive control for amplicon sequencing of microbiotas from the same ecosystem

Full length 16s available





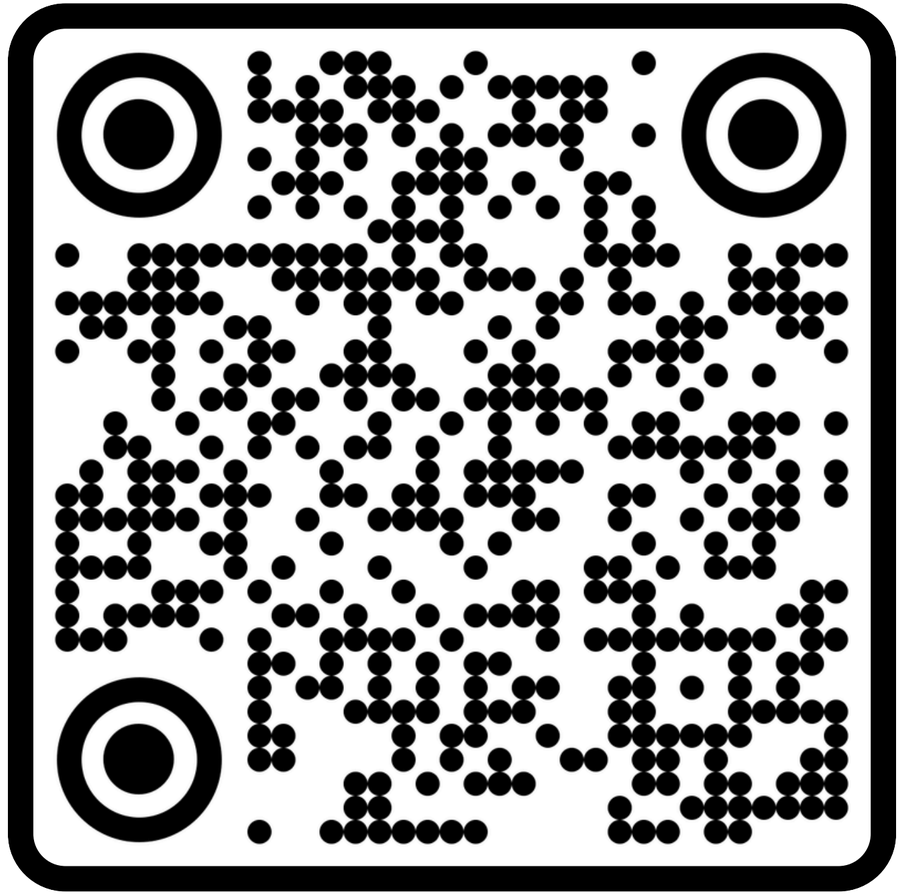
= ~12x increased throughput!

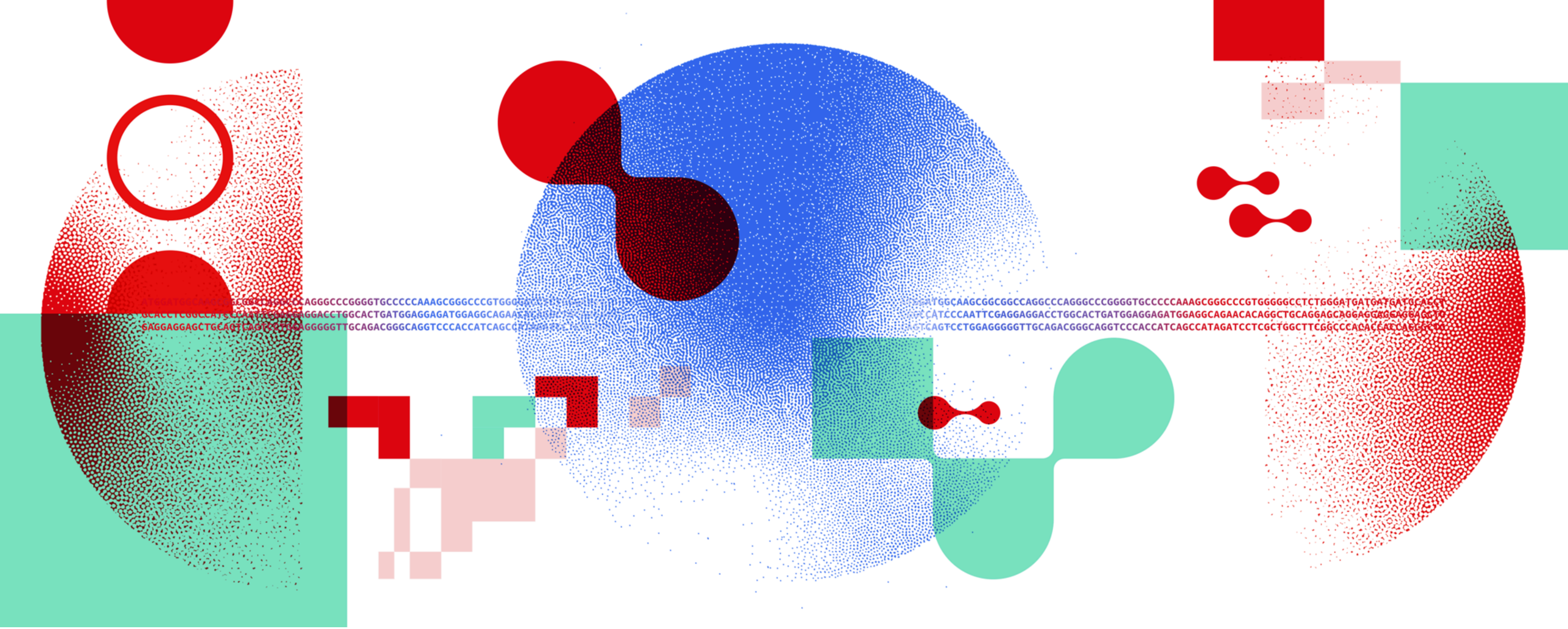


Summary

- **Targeted or amplicon** sequencing is the most used method for microbiome analysis.
- OTUs are defined arbitrarily vs ASVs are defined by the sequence
- Choice of the reference database is critical
- **MOCK sample** helps to validate the pipeline
- **Full length 16S now available!**

Practicals





Thank you

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SIB